3D modelling of confined gastruloid aggregates

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November 2, 2026

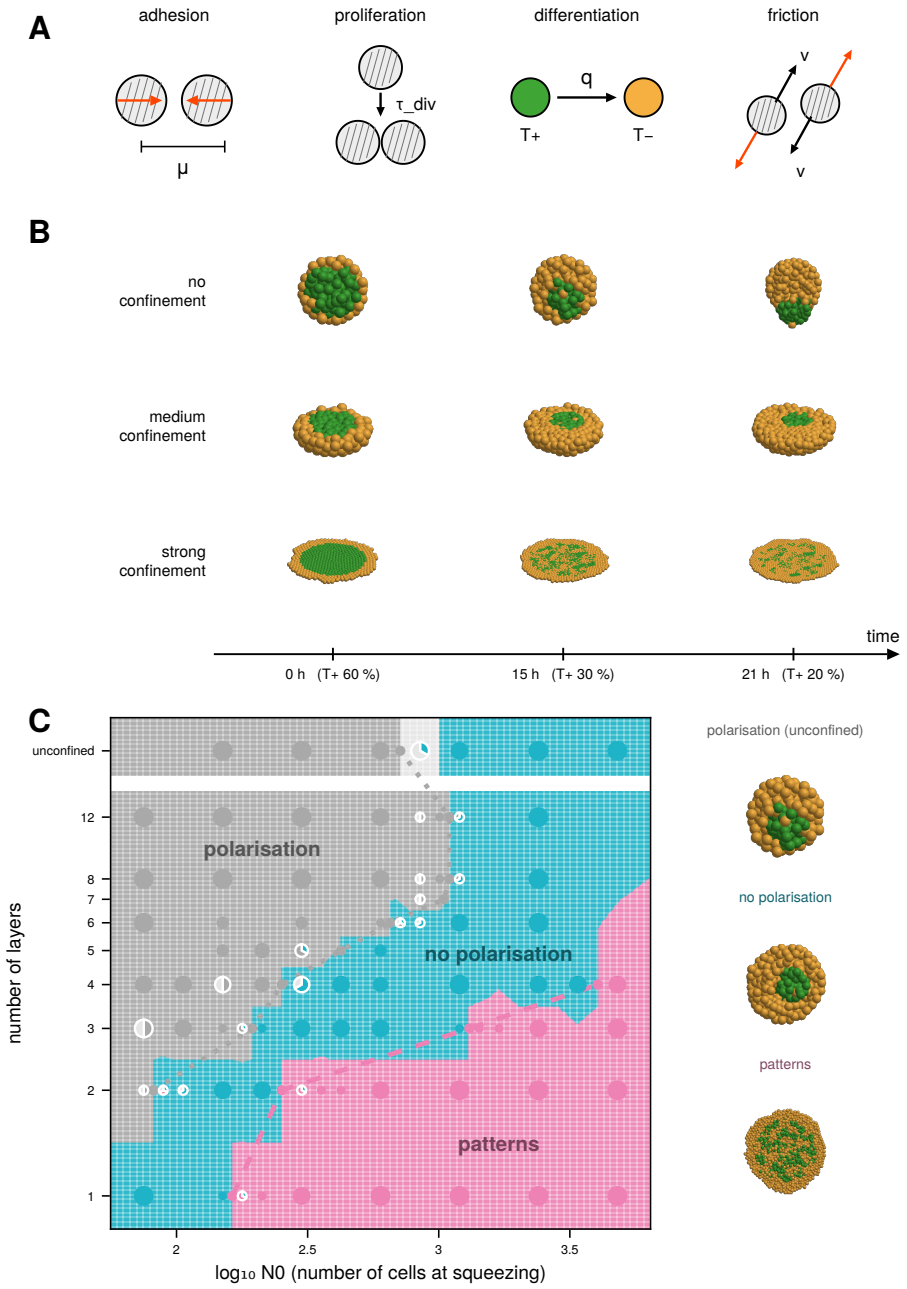


Figure 1: Overview. (A) Model ingredients. (B) Evolution of same aggregate ($N_0 = 600$ cells) without confinement, with medium confinement (4 layers) and with strong confinement (1 layer). (C) Outcome phase space: number of initial cells against the number of layers between the plates.

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1 The experiment

In the expriments, they grow a gastruloid and confine it right before polarisation should occur. Cells transition $A \to B \to C$, and in that moment the distribution is something like: almost no A-cells, a centroid of B-cells, and an outer layer mainly formed of C-cells. On squeezing the aggregate between two plates, there are three possible outcomes.

1. If the confinement is strong enough (i.e. small plate distance), the centroid starts disintegrating usually forming a ring-like structure and then breaks into clusters.
2. For medium confinement, the centroid does not polarise nor breaks into clusters and eventually differentiates away. It may form a ring-like pattern but without breaking.
3. For weak confinement, the centroid polarises as when there is no confinement.

Determining whether confinement is strong or weak is not trivial and is actually key to this research. The experiments observe that the larger the intitial size on squeezing, the most layers we also get clusters with.

The experimentalists image the aggregate and track the cells in order to perform certain measurements.

Spatial cluster quantification. It is used to determine whether clusters have formed.

Spatial clustering was quantified using a density-based clustering approach adapted from DBSCAN (Density-Based Spatial Clustering of Applications with Noise). Clusters were identified based on the local enrichment and spatial connectivity of cells belonging to the cell type B population. For each cell, the fraction of neighbouring cells belonging to the same population was calculated within a defined three-dimensional Euclidean radius of 13 µm, considering all surrounding cells as the reference. Cells for which at least 40% of neighbouring cells belonged to the population of interest were defined as core cells. Core cells were grouped into the same cluster if they were directly or indirectly connected through other core cells within a maximum pairwise distance of 13 µm. Cells belonging to the population of interest that did not fulfil the core-cell criterion were assigned as border cells to the cluster of their nearest core cell, provided that this core cell was located within 13 µm. Only clusters containing at least 20 cells, including core and border cells, were retained for further analysis. Clustering was quantified based on the number of clusters, the fraction of cells assigned to clusters, cluster sizes, and the effective number of clusters (N_{eff}), which accounts for differences in cluster size distribution. The latter was calculated as $N_{\text{eff}} = \frac{1}{\sum p_i^2}$, where p_i represents the fraction of all clustered cells belonging to cluster i, providing a measure of cluster number that accounts for differences in cluster size.

Polarisation. Polarisation is defined as follows. Let r be the center of mass of the aggregate, namely $r = \frac{1}{N}\sum r_i$. Let r_c be the center of mass of the centroid. Let R be the half of the length

of the PC1 axis (longest axis). Then polarisation P is given by

$$P = \frac{|r - r_c|}{R}$$

The cutoff is set to 0.3. To determine it we just looked at the polarisation histogram of control gastruloids. Here is a plot with samples of different sizes. Where you a see that the polarisation degree at that timepoint seems to be also size dependent, but 0.3 includes the majority.

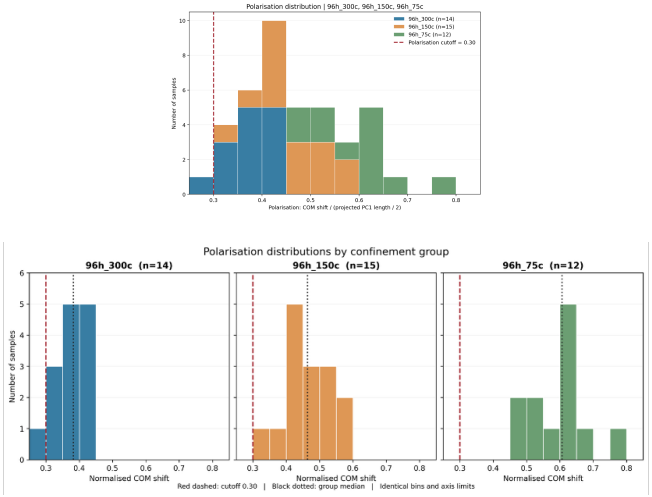


Figure 2: Polarisation histogram of control gastruloids.

2 The physical description

2.1 Simplificating reality

First, we grow and differentiate a gastruloid aggregate before polarisation begins. Instead of the three-state description used in Oriola et al. 2026 ($A \to B \to C$), we assume all A cells have differentiated away at the moment of squeezing and therefore only consider B (T^+) and C (T^-) cells.

We define out initial aggregate to be a spheroid with a central cluster of B cells containing about 60% of the cells in the aggregate. The rest of the cells are in state C. If we were to let this further evolve, this cluster would polarise.

In order to model the behaviour of the confined aggregate, we won't study the confinement process but instead consider an already-squeezed aggregate with a B-centroid. We define it by growing an aggregate under confinement, setting the inital cells as they would result on confining the spherical gastruloid, letting the cells relax, and then starting differentiaion and proliferaiton.

We want to replicate all three behaviours described previously.

Fusion. We use fusion as a sanity check to see that they are not only replicating the confined patterning, just like we do with unconfined polarisation. We see that, under certain conditions, two modelled gasruloids fuse as they do experimentally. In particular, we see that

1. Two spherical B-only aggregates fuse perfectly forming a single sphere.
2. A B-aggregate and a C-aggregate glue together but do not form a single sphere.

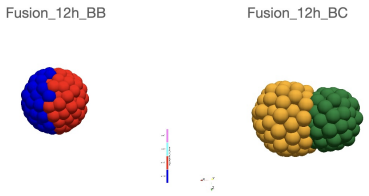


Figure 3: Expected fusion behaviour.

2.2 Modelling 3D aggregates

Mechanics. In the agent-based simulations, a single cell i is represented by the position of its centre of mass $\mathbf{x}_i$ and its dynamics is given by the following overdamped equations of motion:

$$\begin{aligned}\dot{\mathbf{x}}_i &= \mathbf{v}_i\,,\\ \mathbf{v}_i &= \frac{1}{\lambda n_i}\sum_j \mathbf{F}_{ij} + \frac{\alpha}{n_i}\sum_j \mathbf{v}_j\,.\end{aligned}$$

where isotropic friction forces are assumed and $\mathbf{v}_i$ is the cell velocity. λ_s is the friction coefficient for the cell-substrate or cell-ECM interaction, while λ_r is the relative cell-cell friction coefficient.

The passive force $\mathbf{F}_{ij}$ describes cell-cell adhesion and repulsion that depends on the position of the two interacting cells (i,j) as well as their states $(s_i,s_j)\in\{B,C\}^2$:

$$\mathbf{F}_{ij}(s_i,s_j,d_{ij}) = \begin{cases} f(d_{ij})\,\hat{\mathbf{r}}_{ij}F_{\text{rep}}^{(s_i,s_j)}, & \text{if } d_{ij}\le 2r\\ f(d_{ij})\,\hat{\mathbf{r}}_{ij}F_{\text{att}}^{(s_i,s_j)}, & \text{if } 2r<d_{ij}<2\mu r\\ 0, & \text{otherwise}\,,\end{cases}$$

where $F_{\text{rep}}^{(s_i,s_j)}$, $F_{\text{adh}}^{(s_i,s_j)}$ are 3×3 matrices that describe the magnitude of the interaction, $d_{ij}=|\mathbf{x}_i-\mathbf{x}_j|$, $\hat{\mathbf{r}}_{ij}=\frac{\mathbf{x}_i-\mathbf{x}_j}{d_{ij}}$, and

$$f(d_{ij}) = \left(\frac{2r}{d_{ij}}-1\right)\left(\frac{2\mu r}{d_{ij}}-1\right)\,.$$

Remark 1. *In the code we set $\hat{\alpha}=\lambda\alpha$ and so*

$$\mathbf{v}_i = \frac{1}{\lambda n_i}\left(\sum_j \mathbf{F}_{ij} + \alpha\sum_j \mathbf{v}_j\right)\,.$$

Division. Cells undergo division according to the following rules:

- A division axis is randomly chosen over the unit sphere.
- The two daughter cells are generated with equal radius as the parent cell and positioned on opposite sides of the division axis which is centred at the parent cell's location. The daughter cells are generated with their centre of mass at a fixed distance from that of the parent cell.
- A new division time is then assigned to each daughter cell, sampled from

$$\mathcal{U}\big(\tau_{\text{div}}(1-\sigma_{\text{div}}),\ \tau_{\text{div}}(1+\sigma_{\text{div}})\big),$$

Differentiaton. The transition rate from B to C is constant,

$$r_{B\to C}\equiv q\,,$$

so every B cell becomes C with probability $q\,\Delta t$ per time step.

Remark 2. *The fraction of B cells falls from 60% to 20% in about 21 hours, against 24 hours (from 72 to 96 h) in the experiments.*

Confinement. The plates sit at $z = 0$ and $z = L$, with $L = 2.1\,r\times$ (number of layers). They are rigid: a cell touching a plate cannot move further into it, so the slab keeps its thickness during the whole simulation. A cell closer than 0.05 to a plate also feels plate friction, an extra in-plane drag:

$$v_{x,y} = \frac{F_{x,y} + \alpha \sum_j v_{j,(x,y)}}{\lambda n_i + \lambda_{\text{wall}}} .$$

2.3 Analysis of the model

Remark 3. *Pulling two attached cells apart costs the* ***adhesion energy***

$$W_{XY} = \int_{2r}^{2\mu r} |F| \,\mathrm{d}d = F_{att}^{(X,Y)}\, 2r\, A(\mu_{XY}) , \qquad A(\mu) = (1+\mu)\ln\mu - 2(\mu - 1) .$$

These energies set the surface tensions $\sigma_X = W_{XX}/2$ and the interfacial tension $\gamma_{BC} = (W_{BB} + W_{CC})/2 - W_{BC}$. Two aggregates glue without rounding into one sphere when $|\sigma_B - \sigma_C| < \gamma_{BC} < \sigma_B + \sigma_C$.

3 The simulations

The main challenge of the simulations is to replicate all the aforementioned behaviours using the same model. All the parameters are in Table 3. All the figures are three-dimensional; a top view leaves out the cells above the mid-plane, since the outer C cells would hide the B pattern otherwise.

3.1 Initial condition

Figure 4 shows the seeded aggregate. We also tried a diluted core (a quarter of the B cells outside the core), but it makes small squeezed aggregates polarise, so the sharp core is used.

3.2 Fusion

Two aggregates of 150 cells of a single type are placed in contact and followed for 24 hours without division or differentiation (Figure 5). The distance between their centres over the sum of their radii of gyration is about 1.6 when they just touch and 0.5 for a single sphere. BB and CC fuse (0.54 and 0.56), whereas BC glues and keeps two lobes (0.91, close to the 0.95 of Sotiris' reference).

3.3 Polarisation

Unconfined aggregates of up to $N_0 = 600$ cells polarise before B runs out (Figure 6). Larger aggregates ($N_0 = 1200$ to 4800) still have a central cluster when 20% of the cells are B, and the largest never polarise.

3.4 Confinement

Figure 7 follows the same aggregate under no, medium and strong confinement. Figure 8 shows patterns from one to four layers.

In thick slabs the pattern starts as a ring: the core hollows out and then breaks into islands (Figure 9).

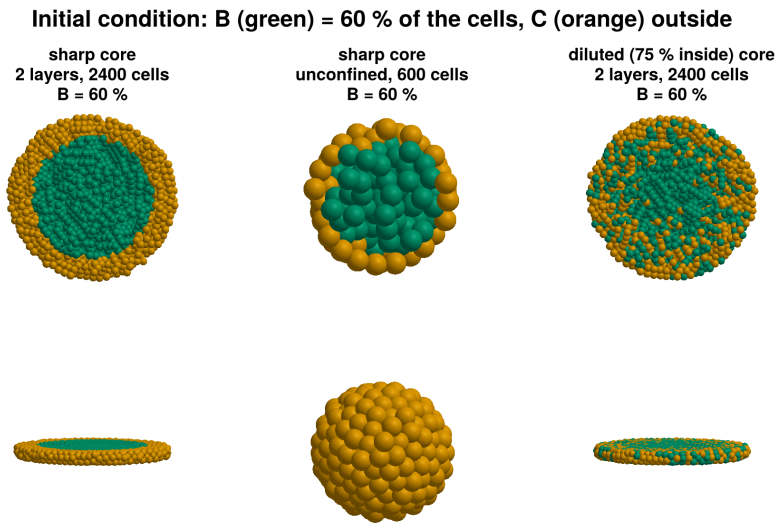


Figure 4: Initial condition: 60% of the cells are B (green), the innermost ones. Left: 2 layers, $N_0 = 2400$. Middle: unconfined, $N_0 = 600$. Right: a diluted core, which is not used.

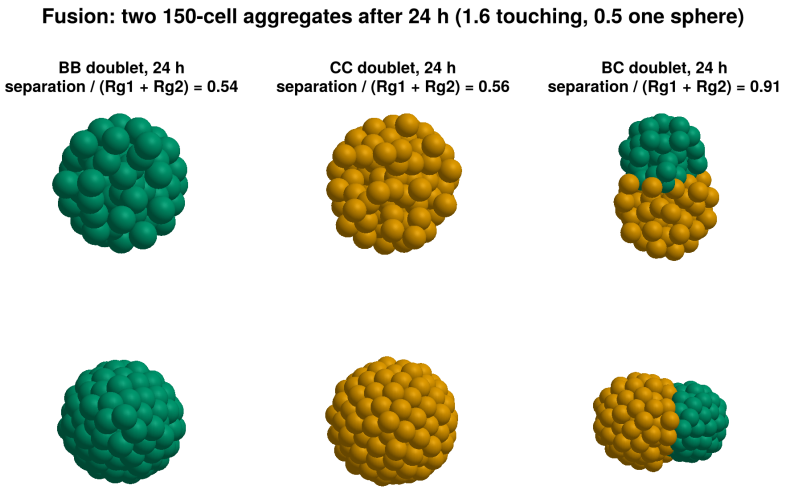


Figure 5: Fusion assay after 24 hours: BB, CC and BC doublets.

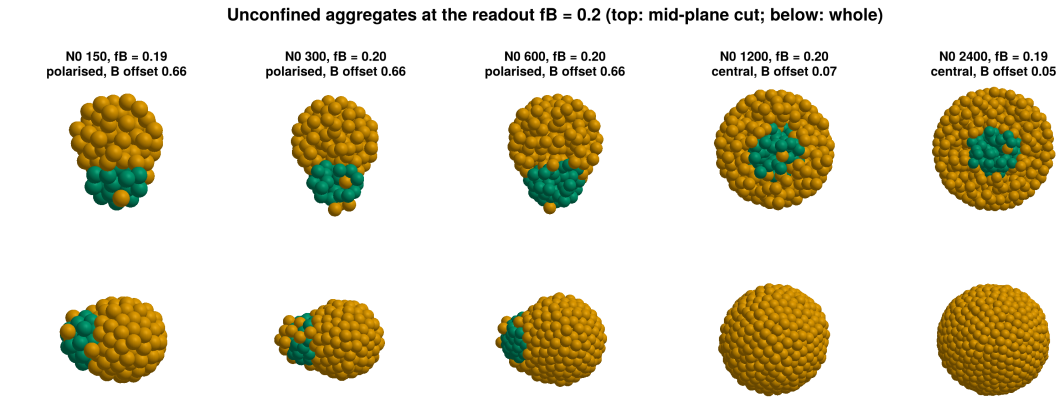


Figure 6: Unconfined aggregates of $N_0 = 150$ to 2400 cells when 20% of the cells are B. Top: cut at the mid-plane; bottom: whole aggregate. Rotated so that B points to the right.

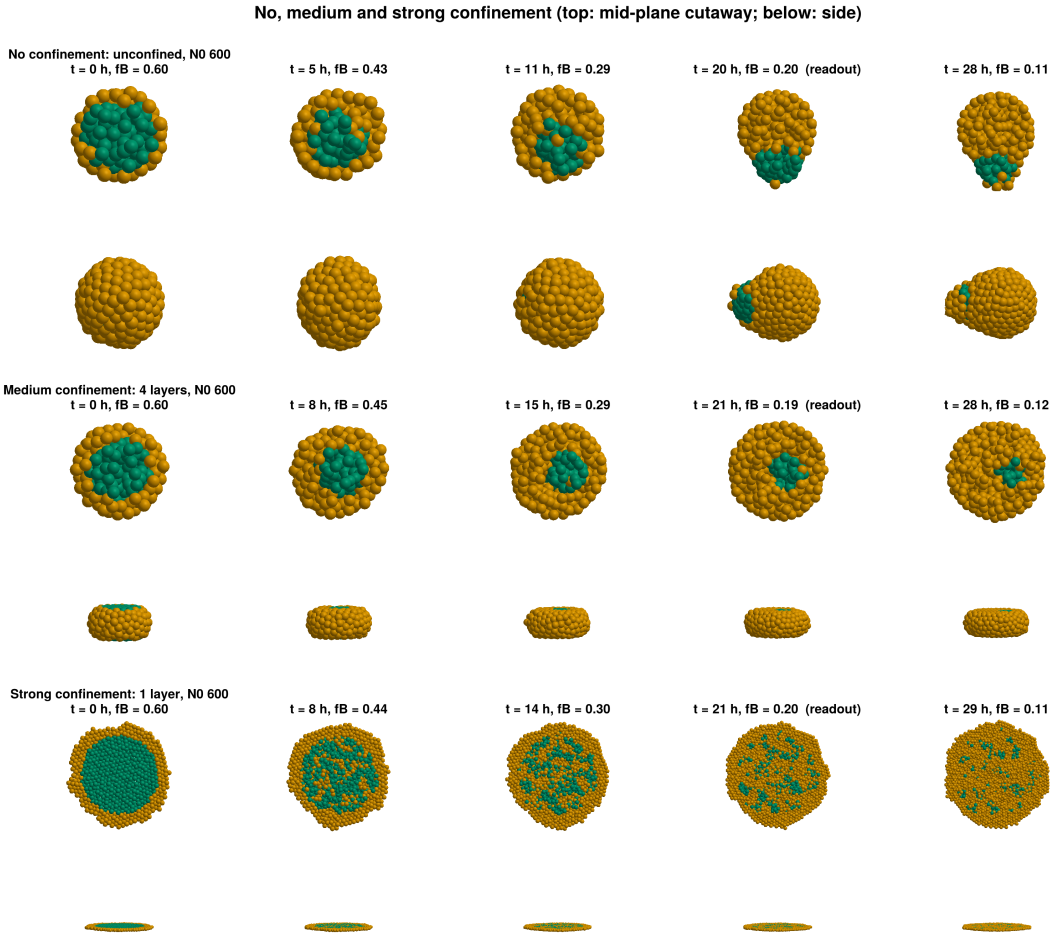


Figure 7: $N_0 = 600$ cells with no confinement (polarises), 4 layers (central cluster that differentiates away) and 1 layer (islands), from the seeding to 12% of B cells. The readout is at 20%.

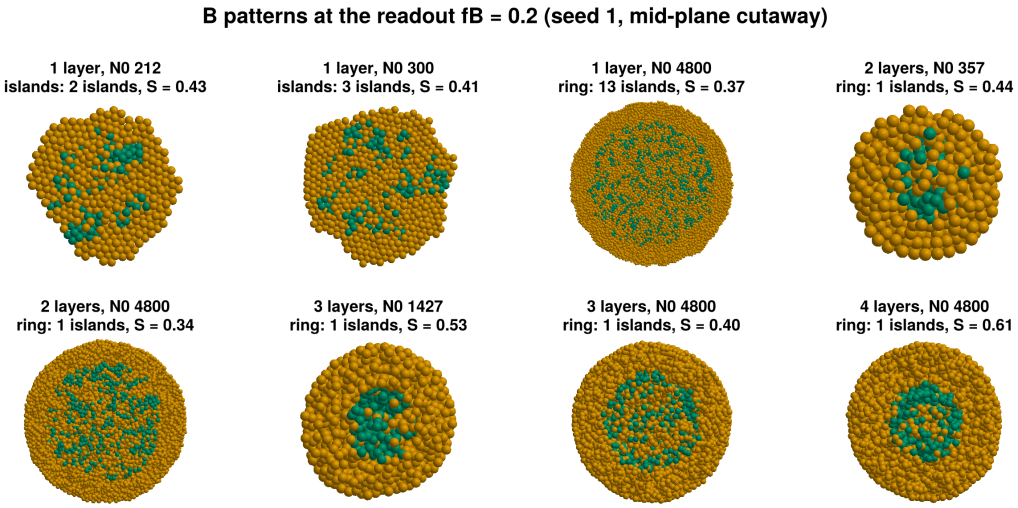


Figure 8: Patterns when 20% of the cells are B, from one to four layers.

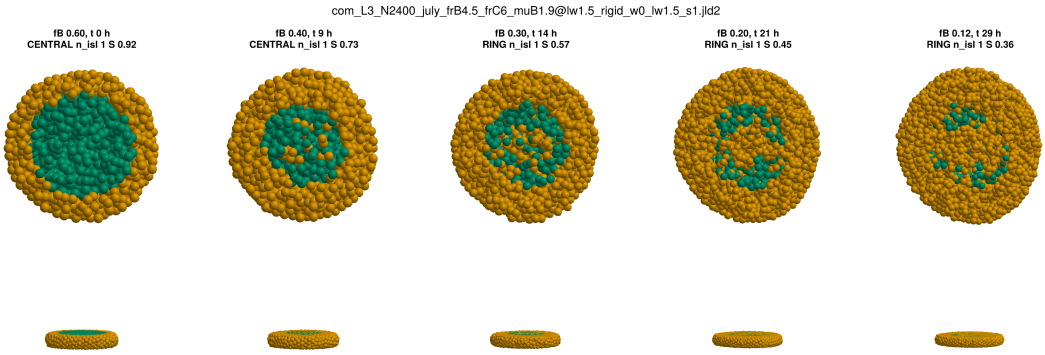


Figure 9: Three layers, $N_0 = 2400$: the central cluster hollows into a ring, which breaks into islands. Top: cut at the mid-plane; bottom: side view.

3.5 Patterns in thick slabs

With the parameters of early October, every slab of three or more layers ended as one compact central cluster, at any N_0 up to 4800. Table 1 lists what we tried at three layers and $N_0 = 2400$. The salt-and-pepper start is the one the July runs used: 75% B, 15% A and 10% C, shuffled inside the core.

Change	**Outcome**	**Looseness S**
B fraction 65% instead of 60%	central	0.88
B cells scattered outside the core (10 to 30% of them)	central	0.86–0.89
C cells mixed into the core (10 to 30% of it, as in July)	central	0.89–0.91
Spring plates (July) instead of rigid ones	central	0.91
Plate friction 5 instead of 1.5	central	0.89
July mechanics ($F_{\text{rep}}^{BB} = 3$, $F_{\text{rep}}^{CC} = 5.1$)	central	0.90–0.93
F_{att}^{CC}: $3 \to 3.6$	central	0.88
F_{att}^{BB}: $6.4 \to 5.6$	central, looser	0.72
λ multiplied by 2, 3 or 4	central (and no polarisation)	0.52–0.84
μ_{BB}: $2 \to 1.85$ or 1.8	B dissolves into C	0.30–0.35
$\boldsymbol{\mu_{BB}}$**: 2 → 1.9**	**ring, then islands**	**0.45**

Table 1: What we tried to obtain patterns at three layers ($N_0 = 2400$). S is about 0.9 for a compact cluster and 0.4–0.6 for loose patterns.

What changes the outcome is the cohesion of the B tissue. In a slab of one or two layers, the C cells that appear inside the B core cannot get out, since cells cannot pass each other, and they split the core. From three layers on, cells pass over and under each other, and a cohesive B core squeezes the new C cells out and stays as one cluster. Shortening the B–B attraction range from $\mu_{BB} = 2$ to 1.9 lowers the B–B adhesion energy from $W_{BB} = 1.02$ to 0.79, still above $W_{CC} = 0.65$. Then small cores stay compact, but a large core in a thick slab opens into a ring. It is the energy and not the range: with $\mu_{BB} = 1.8$ and F_{att}^{BB} raised to keep $W_{BB} = 1.02$, the core stays compact. At $\mu_{BB} = 1.85$ ($W_{BB} \approx W_{CC}$), B is no longer more cohesive than C and simply dissolves into it, even in small aggregates (Figure 10).

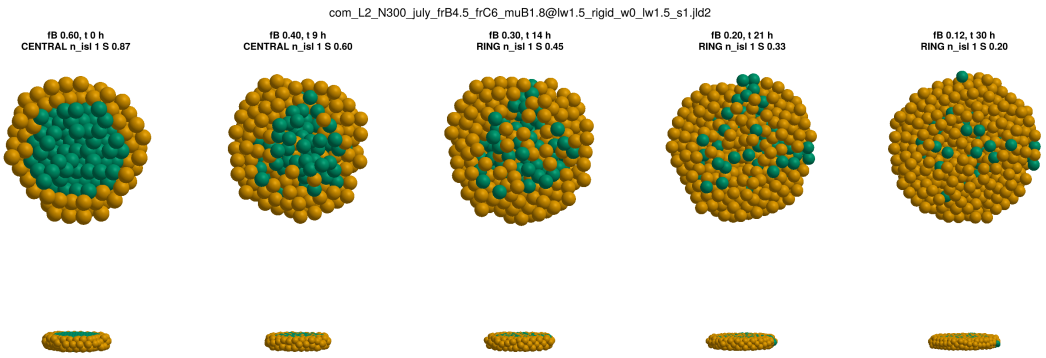


Figure 10: With $\mu_{BB} = 1.8$ the B cells dissolve into the C tissue (2 layers, $N_0 = 300$).

The new parameter set keeps the other behaviours: small squeezed aggregates keep a central cluster, monolayers give sparse islands, unconfined aggregates polarise, and BC fusion glues.

3.6 Phase space

Finally, we compute the phase space relating the outcome, the number of layers and the number of cells N_0 at the moment of squeezing (Figure 11), with 2 or 3 simulations per point (178 in

total). The outcome is read when 20% of the cells are B (96 h in the experiments) and classified as central cluster, pattern (a ring or islands), polarised, or other.

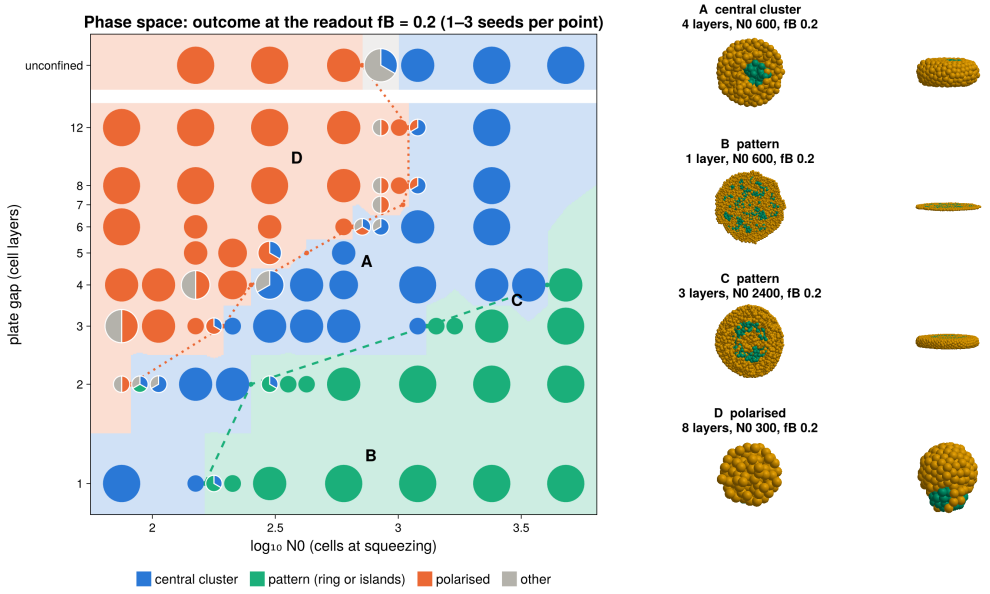


Figure 11: Phase space. Each point is a pie of the outcomes of its simulations; the background is the outcome of the nearest point. Dashed: patterns from here on; dotted: polarised up to here. Right: examples of each outcome.

Layers	**Polarised up to N_0**	**Central cluster**	**Patterns from N_0**
1	–	75–150	≈ 160
2	75	89–212	≈ 250
3	150–178	212–1200	≈ 1300
4	212	300–3394	between 3394 and 4800
6–12	600–1000	up to 2400	none up to 2400
unconfined	600	849–4800	none up to 4800

Table 2: The thresholds of Figure 11.

The threshold for patterns grows quickly with the thickness, from $N_0 \approx 160$ cells at one layer to about 4000 at four layers. Plotted against the diameter of the pancake at the moment of squeezing (Figure 12), the threshold varies much less: patterns appear once the pancake is about 8 to 17 cell diameters wide (roughly 80 to 170 μm). Most of the effect of the thickness is therefore that a thicker slab needs many more cells to reach the same width.

The outcome depends on when it is read. Figure 13 shows the same simulations when only 12% of the cells are B: the rings break into islands, larger aggregates polarise late, and the smallest monolayer patterns shrink back to a central cluster.

4 Open points and possible improvements

1. Large aggregates that are not confined (or confined with a wide gap) neither pattern nor polarise before B runs out. A free B core can always round up, so it does not break into islands.
2. The transition from a central cluster to a ring is gradual: near the threshold the cluster becomes loose and starts opening. Where we put the cut moves the thresholds by about one step in N_0.

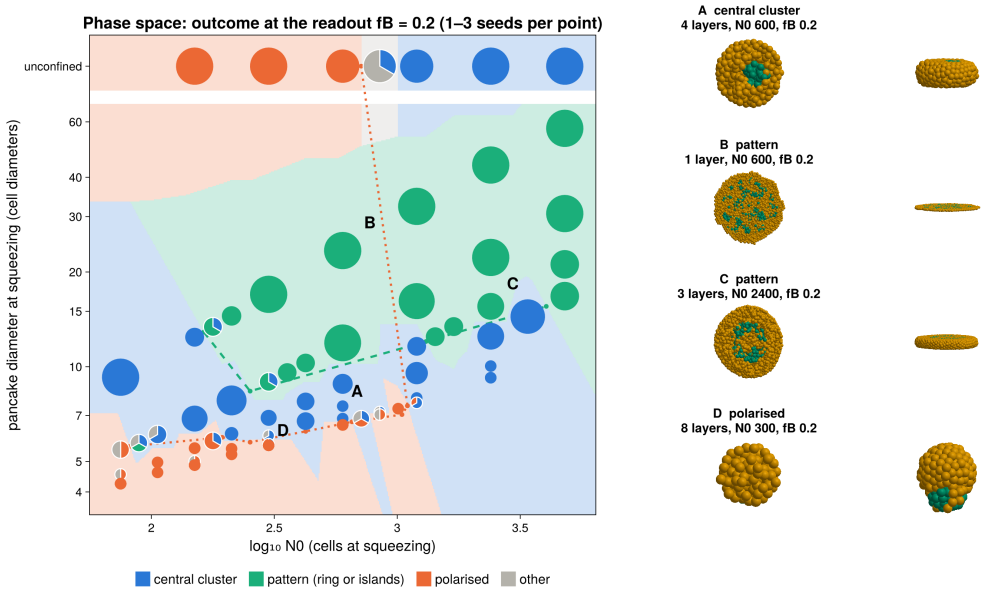


Figure 12: The same phase space against the diameter of the pancake at the moment of squeezing, in cell diameters.

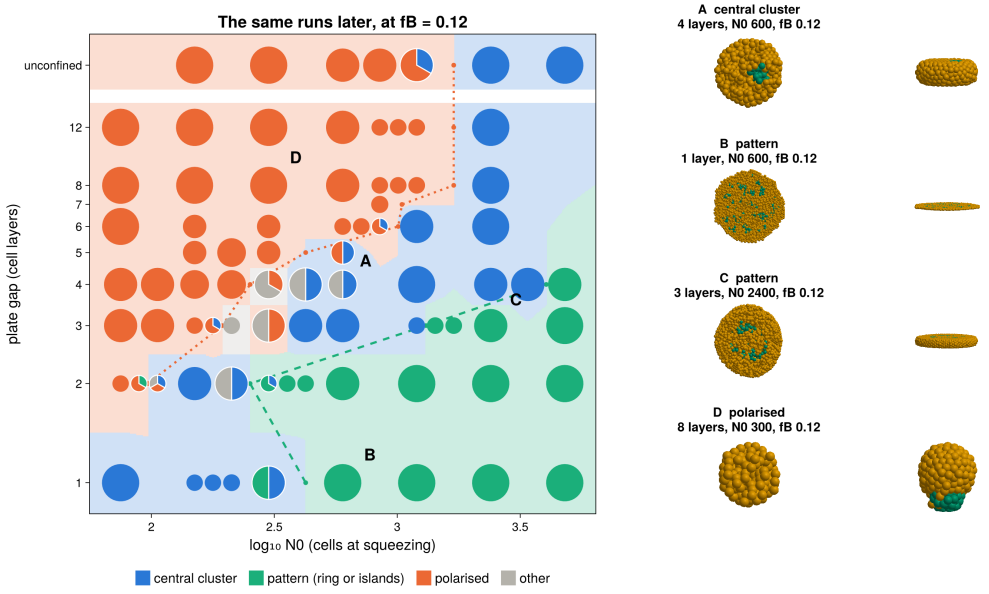


Figure 13: The same simulations, read when 12% of the cells are B.

3. The value $\mu_{BB} = 1.9$ is close to the point where B dissolves into C ($\mu_{BB} = 1.85$).
4. More simulations per point would sharpen the boundaries; the phase space has 2 or 3 per point, and 1 at $N_0 = 4800$.

5 Parameters

Table 3 gives the parameters of the July report (old) and the ones used now (new). Values in red have changed. Entries of the A state are shown as $\cdot$, since A cells no longer exist; in the matrices the order is A, B, C.

Table 3: Parameters of the July report (old) and of this report (new). Values in red have changed.

Symbol	Description	Old (July)	New (October)
Δt	Timestep	0.02	0.015
r	Radius	1	1
	Overlap factor	0.3	0.3
α	Cell-ECM/cell-cell friction parameter	0.9	0.9
τ^A_{div}	Average division time	300	– (no A cells)
τ^B_{div}	Average division time	300	300
τ^C_{div}	Average division time	750	750
σ_{div}	Division time variability	0.5	0.5
λ	Global friction coefficient	1	1
μ	Force range factor	$\begin{pmatrix} 1.55 & 1.5 & 1.525 \\ 1.5 & 1.7 & 1.45 \\ 2.0 & 2.0 & 2.0 \end{pmatrix}$	$\begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & 1.9 & 2 \\ \cdot & 2 & 2.14 \end{pmatrix}$
	Neighbour range factor	1	1
F_{rep}	Repulsion force	$\begin{pmatrix} 3 & 3 & 3 \\ & 3 & 3 \\ & & 5.1 \end{pmatrix}$	$\begin{pmatrix} \cdot & \cdot & \cdot \\ & 4.5 & 3 \\ & & 6 \end{pmatrix}$
F_{att}	Attraction force	$\begin{pmatrix} 6 & 6 & 2.4 \\ & 6 & 1.6 \\ & & 4.8 \end{pmatrix}$	$\begin{pmatrix} \cdot & \cdot & \cdot \\ & 6.4 & 1.6 \\ & & 3 \end{pmatrix}$
p	Fixed transition rate from A to B	0.009333	– (no A cells)
q	Fixed transition rate from B to C	0.004667	0.004667
K	Feedback parameter	0.2	– (no feedback)
λ_{wall}	Plate friction (cells closer than 0.05 to a plate)	–	1.5
	Plates	–	rigid
L	Plate gap	–	$2.1\,r\times$ layers
ϕ^0_B	Initial proportion of B cells	–	0.6 (central core)
	Readout: proportion of B cells	–	0.2
n_i	Number of neighbours		
n_i^X	Number of neighbours in state X		
ϕ_X	Proportion of cells in state X		
T_0	Typical time scale	4 min	4 min
	Time units per hour	–	15
R_0	Typical length scale	5 µm	5 µm
F_0	Typical force scale	20 pN	20 pN

Note on the old values. They are the values in the table of the July report. The July simulations themselves used Sotiris' fusion set,

$$\mu = 2\begin{pmatrix} 1 & 0.91 & 1 \\ 0.91 & 1 & 1 \\ 1 & 1 & 1.07 \end{pmatrix}, \qquad F_{\text{att}} = 4\begin{pmatrix} 0.9 & 1 & 0.6 \\ 1 & 1.6 & 0.4 \\ 0.6 & 0.4 & 0.75 \end{pmatrix},$$

with $F^{BB}_{\text{rep}} = 3$ and $F^{CC}_{\text{rep}} = 5.1$, and Hookean plates. Compared with those, the only changes are F^{BB}_{rep} $(3 \to 4.5)$, F^{CC}_{rep} $(5.1 \to 6)$, μ_{BB} $(2 \to 1.9)$, the plates (rigid, with friction $\lambda_{\text{wall}} = 1.5$) and the removal of the A state. The BC entries ($\mu_{BC} = 2$, $F^{BC}_{\text{att}} = 1.6$, $F^{BC}_{\text{rep}} = 3$) and q are unchanged. The typical time scale $T_0 = 2$ h does not match the 15 time units per hour implied by q; it is worth checking.

References

Oriola, David et al. (2026). "Collective fate decisions and cell rearrangements underlie gastruloid symmetry breaking". In: *bioRxiv*. DOI: 10.1101/2024.12.16.628776. eprint: https://www.biorxiv.org/content/early/2026/02/24/2024.12.16.628776.full.pdf. URL: https://www.biorxiv.org/content/early/2026/02/24/2024.12.16.628776.